

A choice-free Lean formalization of Bishop–Cheng dominated convergence over presented reals

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Author contributions

The paper has a single author, who conceived the work, carried out the formalization, built and ran the audit apparatus, and wrote the manuscript, and who is responsible for all of it.

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Competing interests

The author has no competing interests to declare that are relevant to the content of this article.

Data and code availability

There is no dataset. The Lean 4 artifact, the audit apparatus and the shipped audit logs are archived and citable at `10.5281/zenodo.22333837`, and are also at <https://github.com/turahigashi/choicefree-bc-dct> under tag `v0.7.8`, an annotated tag pointing at the commit the deposit was cut from. Reproducing the build requires Lean 4.30.0 and mathlib 4.30.0; the remaining checks need only Python 3 and `coreutils`.

Ethics approval

Not applicable.

Preprint

Two earlier drafts of this manuscript, under the title “A choice-free Lean formalization of a profile-based dominated convergence theorem in presented Bishop–Cheng measure theory”, were deposited on Zenodo as preprints under CC BY 4.0 on 2026-08-09 (`10.5281/zenodo.21859192`) and 2026-08-13 (`10.5281/zenodo.21911760`); the concept DOI is `10.5281/zenodo.21859191`. Neither has been peer reviewed, and the present manuscript supersedes them. The work has not been posted to arXiv. The Lean 4 artifact it reports, together with the audit apparatus and the shipped audit logs, is archived separately under the MIT licence at `10.5281/zenodo.22333837`.